ANTRIANA SOFRONIOU

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Palaiseau, France Nicosia, Cyprus



EDUCATION

Bachelor of Science

Ecole Polytechnique

2023-2026 Palaiseau, France

Double Bachelor of Mathematics and Physics with a minor in Biology
Activities:

- Performed the Quadrille des Lanciers at 133e Bal de l'X.
- Attended mathematics seminars organized by students and researchers.
- KnittiX Committee president (Organized weekly crochet/knitting meetings).
- Performed with RockX (rock dancing committee) in multiple events.
- Took part in the mathematical crochet project presented during a bachelor event (bachelorgiving).

Apolytirion - High School Diploma

Lykeio Kykkou A - High school

2020-2023 Nicosia, Cyprus

- Main courses: Greek, Mathematics, Physics, Biology, Chemistry
- Grade: 19.56/20

EXPERIENCES

Ontology Mapping and Data Processing of the ITHANET Database

Cyprus Institute of Neurology & Genetics (CING)

July 2026 - August 2026

Worked on the FAIRification and processing of thalassaemia-related data from the ITHANET database through ontology mapping, data standardisation, and validation of genetic variant information. Used R, Python, Ontotext Refine, and biomedical ontologies including SNOMED CT, NCIT, MONDO, and HPO. Supervised by Dr. Petros Kountouris, Dr. Coralia Stephanou and Mrs. Maria Xenophontos

Quantitative assessment of collagen hydrogel stiffness under varying polymerization conditions

LMS - Ecole Polytechnique

September 2025 - December 2025

By preparing collagen gels with different collagen concentrations, we aimed to determine the stiffness of collagen hydrogels prepared in the lab. Supervised by Prof. Jean-Marc Allain

Cryoelectron Microscopy

BIOC - Ecole Polytechnique

June 2025 - July 2025

Conducted protein production, ribosome complex reconstitution, and cryo-EM sample preparation and analysis (ChimeraX, Coot, CryoSparc). Supervised by Prof. Clement Madru

ACHIEVEMENTS

- **Pancyprian Mathematical Contest - 12th grade**
  10th place in the first phase (regional) 12th place in the second phase (national)
- **Pancyprian Biology Olympiad 12th grade**
  38th place in the first phase and 41st place in the second phase

MASTERCLASS

CERN Physics MasterClass — University of Cyprus (March 2023) Participated in a one-day workshop on particle physics and data analysis using real proton–proton collision data from the CMS experiment at CERN.

SKILLS

- Python coding
- ChimeraX, Coot, CryoSparc, VMD, Proteus, SciDavis softwares
- Word, PowerPoint, Excel, Latex

LANGUAGES

- Greek - Mother Tongue
- English - IELTS 7.5 - C1
- French - TCF 400 points - B2
- Spanish - A2

HOBBIES

- Piano (ABRSM Grade 8: Distinction)
- Ballet (RAD Grade 5: Merit), Hip-Hop, French Rock (RockX committee co-president)
- Crochet, Knitting, Sewing (KnittiX presdtent)
- Climbing, Swimming

Computational Biophysics Project

University of Cyprus

July 2024 - August 2024

Protein Design using Proteus and Visual Molecular Dynamics (VMD) software. The goal was to increase the binding affinity of AnkB with GABARAP using Monte-Carlo methods through the Proteus software and then visualizing it using VMD. Supervised by Prof. Georgios Archontis and Prof. Savvas Polydorides

Mathematical crochet project

Ecole Polytechnique

September 2025 - November 2025

Illustrated mathematical objects using crochet. The creations where presented during Bachelorgiving.

BACHELOR THESIS

Study of the function of the uS2 protein

Ecole Polytechnique - BIOC

December 2025

I joined a research team studying translation initiation using techniques such as cryo-electron microscopy. Preliminary results suggested that the uS2 protein plays a role in stabilizing the initiation complex, and my goal was to determine the dissociation constant of uS2 with the 30S ribosomal subunit. During this project, I performed site-directed mutagenesis, DNA amplification and purification, protein expression and purification, and fluorescence anisotropy experiments. Although the selected mutations appeared promising, the dissociation constant could not be determined. The project is ongoing as part of my semester 6 biology course, where I will explore alternative mutations. Supervised by Prof. Emmanuelle Schmitt, Prof. Clément Madru, and PhD student Valentina Varriale.